

untitled0

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```
[1]: from google.colab import files
      uploaded = files.upload()
```

<IPython.core.display.HTML object>

Saving 01 Retail Inventory.xlsx to 01 Retail Inventory.xlsx

import pandas as pd import numpy as np import matplotlib.pyplot as plt import seaborn as sns

```
[2]: import pandas as pd
      import numpy as np
      import matplotlib.pyplot as plt
      import seaborn as sns
```

1 Task 1 – Data Understanding

1.1 Load Dataset

```
[3]: df = pd.read_excel("01 Retail Inventory.xlsx")

      df.head()
```

```
[3]:
```

	Product ID	Product Name	Category	Supplier Name \
0	PRD-1001	Wine Glasses	Home & Kitchen	Heritage Brands
1	PRD-1002	Eye Mask	Beauty	Coastal Trading
2	PRD-1003	Resistance Band	Sports & Outdoors	National Supply Group
3	PRD-1004	Action Figure	Toys & Games	Pacific Wholesale
4	PRD-1005	Serving Tray	Home & Kitchen	Coastal Trading

	Unit Cost	Retail Price	Opening Stock	Units Sold	Closing Stock \
0	23.78	38.89	135	118	17
1	15.67	24.53	86	10	76
2	5.59	11.88	319	195	124
3	9.72	18.01	173	140	33
4	8.22	13.90	32	8	24

	Reorder Point	Stock Status
0	30	Low Stock

1	9	In Stock
2	47	In Stock
3	39	Low Stock
4	7	In Stock

1.2 Dataset Information

```
[4]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 11 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Product ID            1000 non-null   object
1   Product Name          1000 non-null   object
2   Category              1000 non-null   object
3   Supplier Name         1000 non-null   object
4   Unit Cost             1000 non-null   float64
5   Retail Price          1000 non-null   float64
6   Opening Stock         1000 non-null   int64
7   Units Sold            1000 non-null   int64
8   Closing Stock         1000 non-null   int64
9   Reorder Point         1000 non-null   int64
10  Stock Status          1000 non-null   object
dtypes: float64(2), int64(4), object(5)
memory usage: 86.1+ KB
```

1.3 Missing Values Check

```
[5]: df.isnull().sum()
```

```
[5]: Product ID      0
     Product Name    0
     Category        0
     Supplier Name    0
     Unit Cost        0
     Retail Price     0
     Opening Stock    0
     Units Sold       0
     Closing Stock    0
     Reorder Point    0
     Stock Status     0
     dtype: int64
```

CONCLUSION : The dataset contains 1000 inventory records and 11 attributes. All columns have complete data with no missing values. The dataset includes both numerical and categorical variables, making it suitable for inventory analysis and further exploratory data analysis.

2 TASK 2 Data Cleaning

2.1 Duplicate Check

```
[6]: df.duplicated().sum()
```

```
[6]: np.int64(0)
```

2.2 Missing Values

```
[7]: df.isnull().sum()
```

```
[7]: Product ID      0
     Product Name   0
     Category       0
     Supplier Name  0
     Unit Cost      0
     Retail Price   0
     Opening Stock  0
     Units Sold     0
     Closing Stock  0
     Reorder Point  0
     Stock Status   0
     dtype: int64
```

2.3 Stock Validation

```
[8]: df[['Opening Stock', 'Units Sold', 'Closing Stock', 'Reorder Point']].describe()
```

```
[8]:
```

	Opening Stock	Units Sold	Closing Stock	Reorder Point
count	1000.000000	1000.000000	1000.000000	1000.000000
mean	262.451000	137.040000	125.411000	44.98300
std	140.104457	103.623232	99.023521	27.54468
min	20.000000	3.000000	4.000000	2.00000
25%	141.750000	53.000000	44.000000	22.00000
50%	261.000000	108.500000	96.000000	41.00000
75%	388.000000	204.000000	190.000000	64.25000
max	499.000000	463.000000	432.000000	119.00000

CONCLUSION : No duplicate records were found in the dataset. No missing values were detected, so no data cleaning was required. Validation of stock-related fields showed that all values are positive and within reasonable ranges. Therefore, the dataset is clean and ready for analysis.

3 Task 3 – Exploratory Data Analysis

3.1 Category-wise Product Count

```
[9]: df['Category'].value_counts()
```

```
[9]: Category
      Beauty          134
      Sports & Outdoors 132
      Electronics      126
      Office Supplies  126
      Home & Kitchen   124
      Toys & Games     123
      Grocery          119
      Clothing         116
      Name: count, dtype: int64
```

3.2 Category-wise Units Sold

```
[10]: df.groupby('Category')['Units Sold'].sum().sort_values(ascending=False)
```

```
[10]: Category
      Toys & Games      19704
      Home & Kitchen    18387
      Beauty           18137
      Electronics      17621
      Office Supplies   17112
      Sports & Outdoors  16979
      Grocery          15364
      Clothing         13736
      Name: Units Sold, dtype: int64
```

3.3 Top 10 Best Selling Products

```
[11]: top10 = df.sort_values('Units Sold', ascending=False)

      top10[['Product Name','Units Sold']].head(10)
```

```
[11]:      Product Name  Units Sold
      721  Water Bottle      463
      476  Canned Soup      456
      791  Coffee Mug Set    435
      673  LED Desk Lamp    432
      810  Coloring Set     428
      243  Notebook A5      423
      970  Baseball Cap     422
      550  Whistle          422
      931  Rice Crackers    417
      469  Building Blocks   416
```

3.4 Top 10 Low Stock Products

```
[12]: lowstock = df.sort_values('Closing Stock')
```

```
lowstock[['Product Name','Closing Stock']].head(10)
```

```
[12]:
```

	Product Name	Closing Stock
45	Basic T-Shirt	4
385	Coloring Set	4
819	Notebook A5	4
654	Storage Container	4
425	Nail File Set	5
391	Eye Mask	5
703	Coffee Mug Set	5
977	Cotton Pads	6
698	Whistle	6
709	Dry Shampoo	6

3.5 Supplier-wise Inventory Distribution

```
[13]: df['Supplier Name'].value_counts()
```

```
[13]: Supplier Name
Premier Merchandise      119
National Supply Group    116
Heritage Brands           109
Coastal Trading           103
Global Source Direct      103
United Imports            99
Pacific Wholesale         98
Summit Wholesale Inc       92
Metro Distributors        89
AllGoods Supply Co        72
Name: count, dtype: int64
```

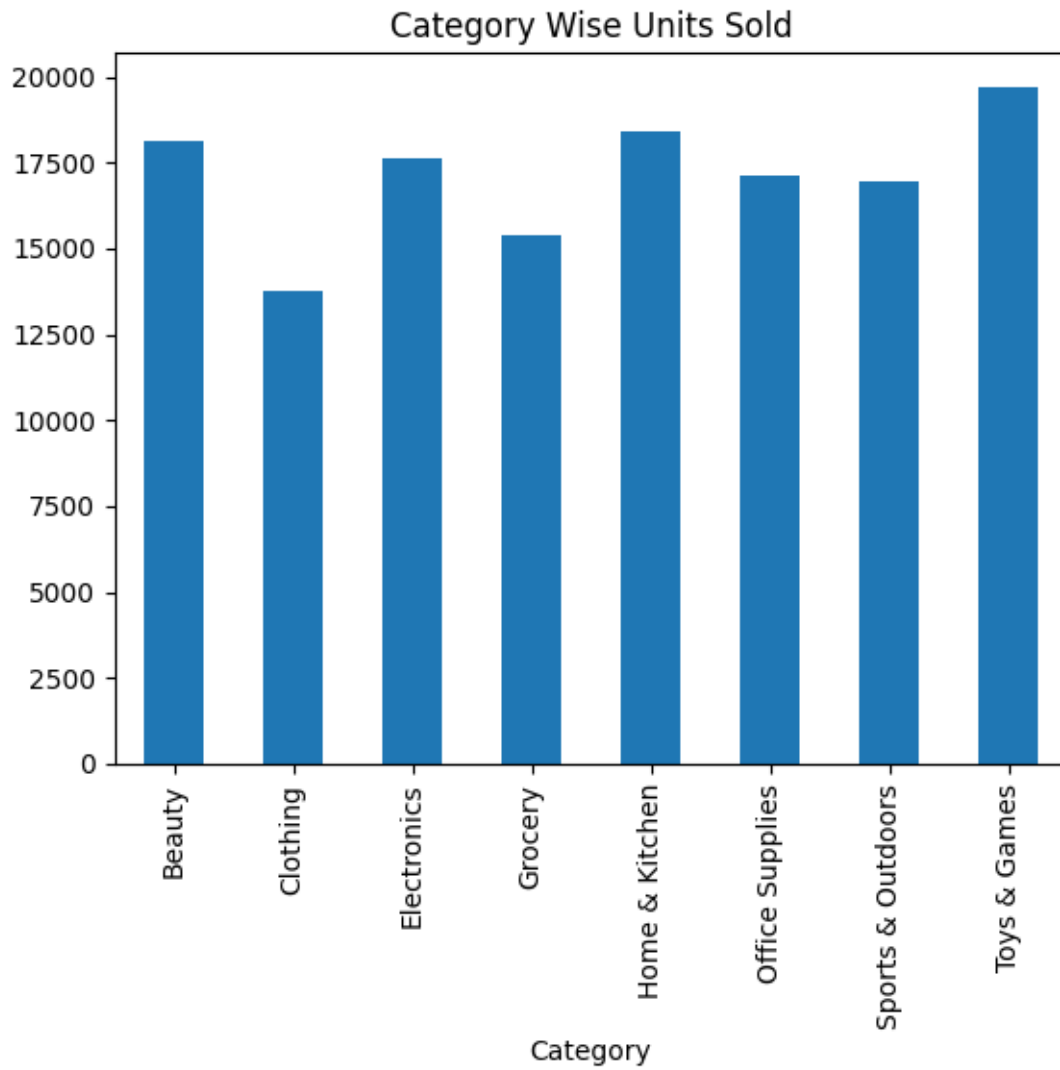
CONCLUSION: The analysis identified the most popular product categories, best-selling products, low-stock items, and supplier contributions. The results provide valuable insights into inventory performance and help in making stock replenishment and supplier management decisions.

4 Task 4 – Charts

4.1 Bar Chart

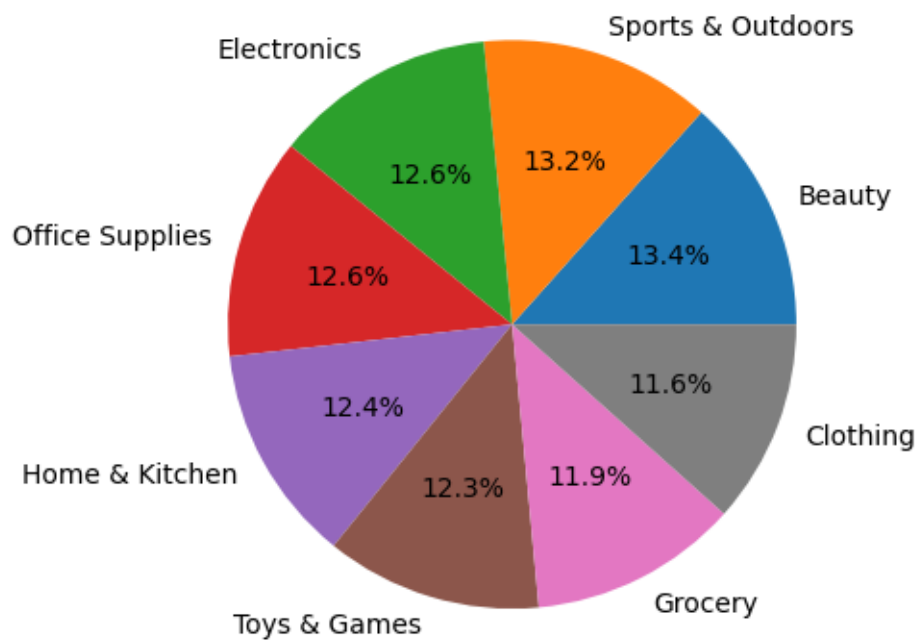
```
[14]: category_sales = df.groupby('Category')['Units Sold'].sum()

category_sales.plot(kind='bar')
plt.title("Category Wise Units Sold")
plt.show()
```



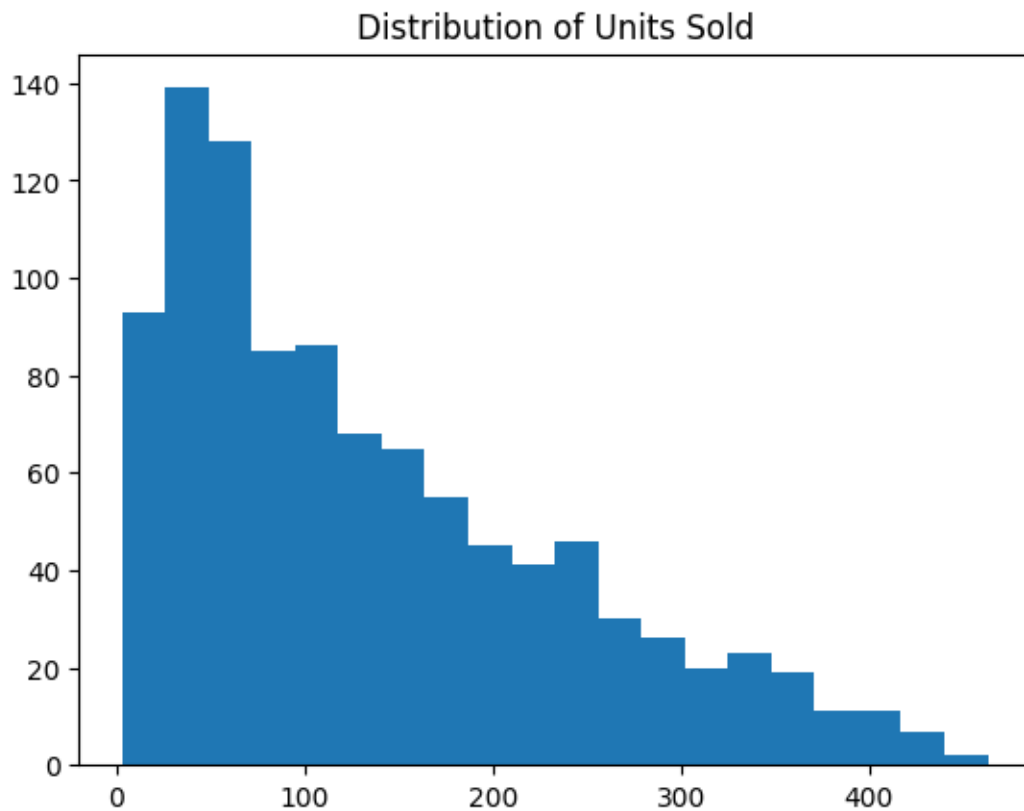
4.2 Pie Chart

```
[15]: df['Category'].value_counts().plot(  
      kind='pie',  
      autopct='%1.1f%%'  
      )  
  
plt.ylabel('')  
plt.show()
```



4.3 Histogram

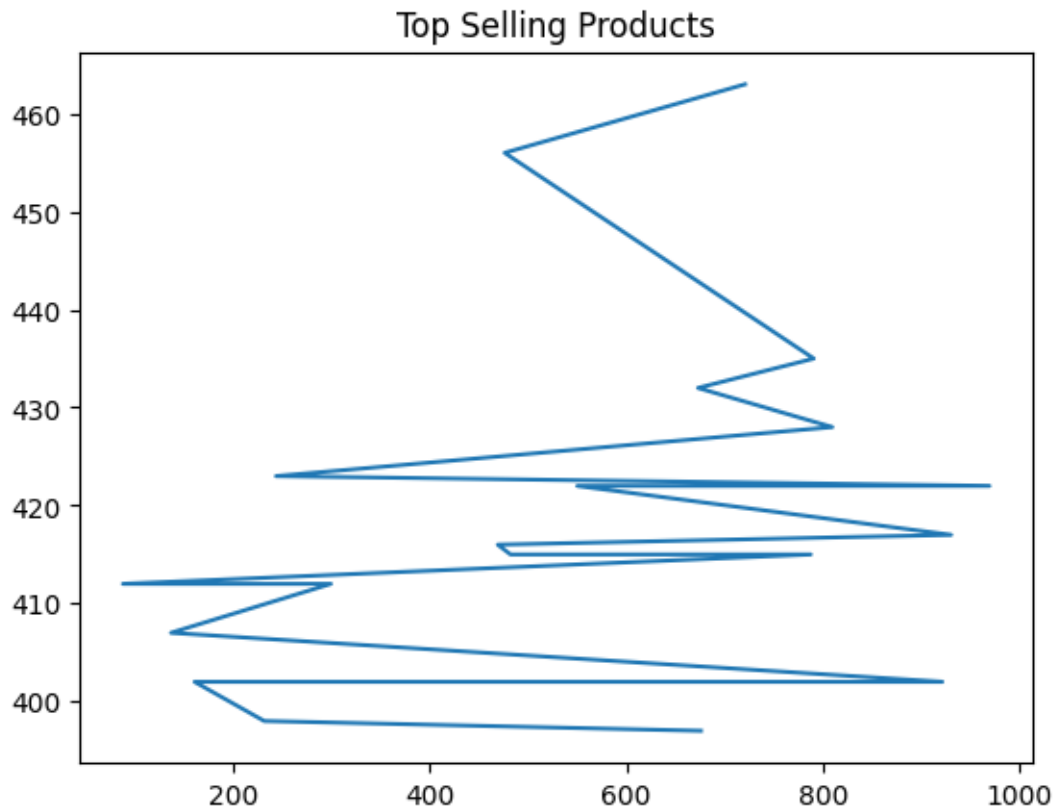
```
[16]: plt.hist(df['Units Sold'], bins=20)
plt.title("Distribution of Units Sold")
plt.show()
```



4.4 Line Graph

```
[17]: top = df.sort_values('Units Sold', ascending=False).head(20)

plt.plot(top['Units Sold'])
plt.title("Top Selling Products")
plt.show()
```

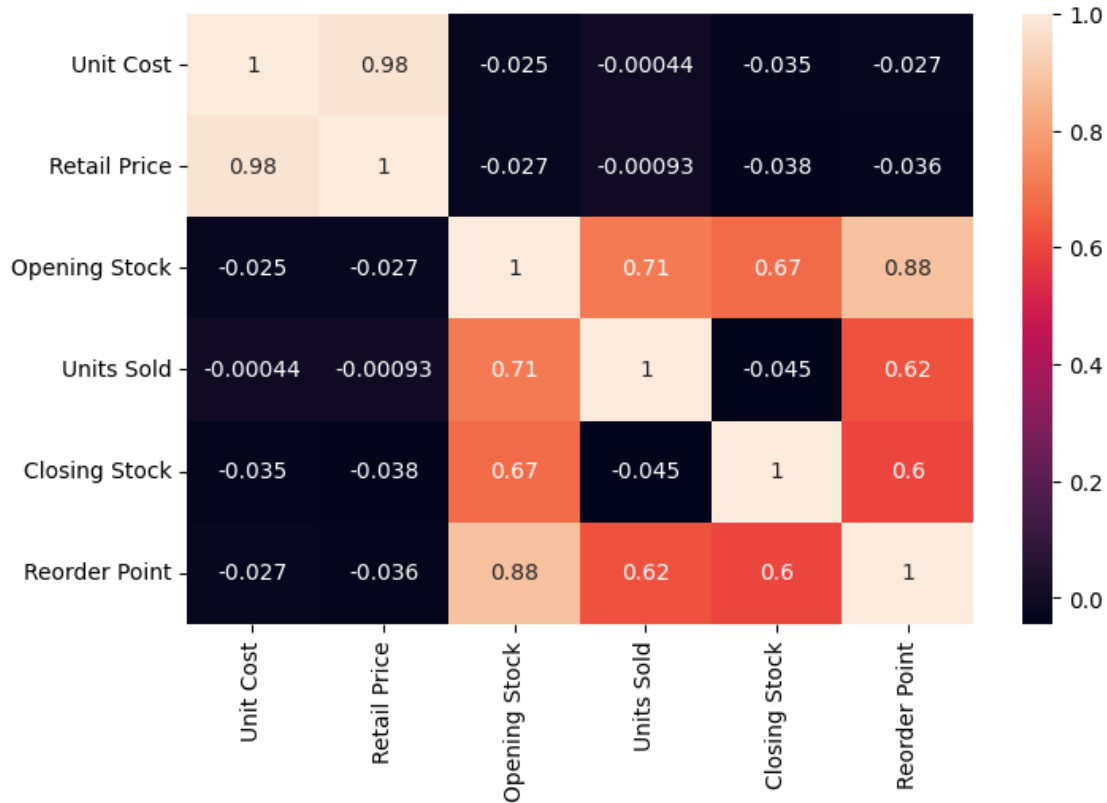


4.5 Heatmap (Bonus)

```
[18]: plt.figure(figsize=(8,5))

sns.heatmap(
    df[['Unit Cost',
        'Retail Price',
        'Opening Stock',
        'Units Sold',
        'Closing Stock',
        'Reorder Point']].corr(),
    annot=True
)

plt.show()
```



CONCLUSION: Data visualizations helped in understanding sales trends, stock distribution, category performance, and relationships between inventory variables. The charts provide a clear and effective representation of inventory data, supporting better business decision-making.

5 Task 5 – Business Insights

5.0.1 5.1 Products Requiring Immediate Replenishment

Products with the lowest closing stock levels are approaching reorder points and should be restocked immediately to avoid stock shortages.

5.0.2 5.2 Fast-Moving Product Categories

Categories with the highest units sold demonstrate strong customer demand and require frequent inventory monitoring.

5.0.3 5.3 Relationship Between Stock and Sales

Products with higher sales generally show lower closing stock levels, indicating that demand has a direct impact on inventory availability.

5.0.4 5.4 Supplier Contribution Analysis

A few suppliers contribute a significant portion of the inventory, indicating dependency on limited suppliers.

5.0.5 5.5 Inventory Optimization Opportunity

Slow-moving products remain in stock for longer periods and may require promotional strategies or adjusted purchasing decisions.

5.0.6 5.6 Business Recommendation

The company should implement automatic reorder alerts, regularly monitor fast-selling products, and diversify suppliers to improve inventory management efficiency.

6 Final Conclusion

The Retail Inventory Analytics project successfully analyzed inventory data using Python and Google Colab. Data cleaning confirmed that the dataset contained no missing values or duplicate records. Exploratory analysis and visualizations helped identify sales trends, inventory patterns, supplier contributions, and low-stock products. The insights generated from this analysis can support better inventory planning, supplier management, and business decision-making.